

CHANGE REQUEST

CR-03

Event Lead Capture & YAYASAN PENERAJU Course Registration Module

Base SRS: Sulfri Trainer Portal — Personal Brand Edition v1.0
Amendment extends: CR-01 (Badge Wallet) + CR-02 (Skill Tree) where applicable
Target Version: v1.3

1. Executive Summary

This Change Request introduces a QR-Driven Event Lead Capture and Course Registration Module for Mohd Sulfri Mohd Harris's Trainer Portal. The module enables Sulfri to deploy a QR code at physical or virtual events that routes participants to a structured registration flow where they:

- Submit their personal details via a mobile-optimised form
- Select one or more courses they are interested in attending
- Are informed that all available courses are offered under YAYASAN PENERAJU

On the admin side, all submitted leads are centrally stored, exportable, and linked to courses. Admins can manage the course catalogue (add, edit, hide, or retire courses) and track interest per course over time.

This module is fully aligned with the admin-controlled content philosophy and Next.js + PostgreSQL architecture defined in SRS v1.0. It does not alter the authentication model, user roles, or deployment infrastructure.

2. Alignment with Existing SRS v1.0, CR-01, and CR-02

The original SRS defines a public-facing trainer portal with admin CRUD control over all content. CR-01 extended this with Credly badge management. CR-02 added a Training Expertise Skill Tree. This CR:

- DOES NOT introduce new user roles
- DOES NOT change the authentication model
- DOES NOT alter deployment architecture
- EXTENDS the admin backoffice with a Lead Management and Course Catalogue section
- ADDS a new public-facing QR landing page and registration form

The Event Registration Form is treated as another admin-managed public page, analogous to `/join/{join_code}` in the base SRS, but scoped to event-driven interest capture rather than class enrolment.

3. Scope of Change

3.1 In Scope

Public Pages (New):

- `/event-register` — QR Landing Page & Registration Form
- `/event-register/success` — Confirmation Page

Admin Pages (Extension of Existing Backoffice):

- `/admin/leads` — View, filter, search, and export event leads
- `/admin/courses` — Course catalogue management (CRUD)
- `/admin/qr` — QR code generator and management

3.2 Out of Scope (Future Phase)

- Automated email confirmation to registrants
- Payment integration for course deposits
- Multi-event tracking (single QR endpoint for now)
- Student login portal

4. Functional Requirements (Extension to Section 5 of SRS v1.0)

4.1 QR Code Landing Page

The public-facing QR destination must be a polished, mobile-first page that immediately establishes context and trust.

FR ID	Requirement
FR-63	The system must provide a dedicated public URL (<code>/event-register</code>) that is the target destination for all event QR codes.
FR-64	The landing page must display: trainer name (Mohd Sulfri Mohd Harris), a brief contextual tagline (e.g., 'Register your interest in our upcoming YAYASAN PENERAJU programmes'), and the YAYASAN PENERAJU branding notice.
FR-65	The page must be fully responsive and optimised for mobile devices (primary access via QR scan).
FR-66	The page must load in under 2 seconds on a standard mobile connection.

4.2 Registration Form — Personal Details

FR ID	Requirement
FR-67	The registration form must collect the following personal details: Full Name (required), Email Address (required), Phone Number (required), Organisation / Company (optional), Job Title / Role (optional).
FR-68	All required fields must be validated client-side and server-side before submission is accepted.
FR-69	Email format must be validated. Phone number must be validated for Malaysian format (optional enforcement, configurable by admin).
FR-70	The form must include a consent checkbox: 'I consent to my details being used for course registration and follow-up communications by Mohd Sulfri Mohd Harris.' Submission must be blocked if consent is not given.
FR-71	The system must prevent duplicate submissions using the same email address within a configurable cooldown period (default: 24 hours). A user-friendly message must be shown for duplicate attempts.

4.3 Course Selection

FR ID	Requirement
FR-72	The registration form must display a list of available courses (sourced from the admin-managed course catalogue) as a multi-select checklist or card-based selector.
FR-73	Each course entry must show: Course Title, Short Description (optional), Delivery Mode (online / physical / hybrid), and a YAYASAN PENERAJU badge/label.
FR-74	The participant must be able to select one or more courses. At least one course selection must be required for form submission.
FR-75	If no courses are currently published, the form must display a message: 'Upcoming courses will be announced soon. Submit your details to be notified.' and allow submission without course selection in this case only.
FR-76	The course list must only display courses with status = PUBLISHED and visibility = PUBLIC.

4.4 YAYASAN PENERAJU Notice

FR ID	Requirement
FR-77	The registration page must prominently display a notice stating: 'All courses listed are offered under the YAYASAN PENERAJU programme.' This notice must appear above the course selection section.

FR-78	Admin must be able to customise the YAYASAN PENERAJU notice text via a settings field in the backoffice (stored in profile_settings or a new event_settings table).
FR-79	The YAYASAN PENERAJU name and programme context must not be removable — only the supporting description text is customisable by admin.

4.5 Confirmation Page

FR ID	Requirement
FR-80	Upon successful form submission, the participant must be redirected to /event-register/success.
FR-81	The success page must display: a thank-you message, a summary of selected courses, and the trainer's contact information for follow-up.
FR-82	The success page must invite participants to scan the QR code again if registering on behalf of someone else.

5. Admin Module Requirements

5.1 Course Catalogue Management

FR ID	Requirement
FR-83	Admin must be able to create a new course entry with the following fields: Title (required), Short Description, Full Description, Delivery Mode (online/physical/hybrid), Start Date (optional), End Date (optional), Location (optional), Status (DRAFT / PUBLISHED / COMPLETED / RETIRED), Visibility (PUBLIC / HIDDEN).
FR-84	Admin must be able to edit any existing course.
FR-85	Admin must be able to change a course's status and visibility independently.
FR-86	Admin must be able to archive (RETIRED) a course without deleting it, preserving historical lead linkage.
FR-87	Admin must see a count of interested registrants per course (derived from lead_course_selections table).
FR-88	Admin must be able to reorder courses to control display order on the public registration form.

5.2 Lead Management

FR ID	Requirement
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FR-89	Admin must be able to view all submitted event leads in a paginated table showing: Full Name, Email, Phone, Organisation, Registration Date/Time, and Courses Selected.
FR-90	Admin must be able to search leads by name, email, or organisation.
FR-91	Admin must be able to filter leads by: course selected, registration date range, and consent status.
FR-92	Admin must be able to export leads to CSV. Export must include all fields plus the list of selected course titles.
FR-93	Admin must be able to view a lead detail page showing full submitted information and all course selections.
FR-94	Admin must be able to mark a lead as 'Contacted' or add an internal note (admin-only field, never shown publicly).
FR-95	Admin must NOT be able to permanently delete a lead. Archiving must be supported instead.

5.3 QR Code Management

FR ID	Requirement
FR-96	Admin must be able to generate a QR code pointing to /event-register from the backoffice.
FR-97	The QR code must be downloadable as PNG and SVG formats.
FR-98	Admin must be able to add a UTM source tag (e.g., event name/date) to the QR URL for tracking lead sources (stored as utm_source on the lead record).
FR-99	Multiple QR codes may be generated for different events, each with a unique utm_source tag.

6. Database Amendment (Extension to Section 6 of SRS v1.0)

The following new tables are added. They follow the same naming, audit field, and relationship conventions as tables defined in SRS v1.0.

6.1 event_courses

Field	Type / Constraint	Description
id	PK (UUID)	Primary key
title	VARCHAR (required)	Course title displayed on form
short_description	TEXT (nullable)	Brief descriptor shown on selection card

full_description	TEXT (nullable)	Detailed course description
delivery_mode	ENUM (online/physical/hybrid)	Delivery format
start_date	DATE (nullable)	Scheduled start date
end_date	DATE (nullable)	Scheduled end date
location	VARCHAR (nullable)	Physical location if applicable
status	ENUM (DRAFT/PUBLISHED/COMPLETED/RETIRED)	Lifecycle status
visibility	ENUM (PUBLIC/HIDDEN)	Public display control
display_order	INT	Admin-controlled sort order
created_at	TIMESTAMP	Audit: record creation
updated_at	TIMESTAMP	Audit: last modification

6.2 event_leads

Field	Type / Constraint	Description
id	PK (UUID)	Primary key
full_name	VARCHAR (required)	Participant's full name
email	VARCHAR (required)	Email address
phone	VARCHAR (required)	Phone number
organisation	VARCHAR (nullable)	Company or institution
job_title	VARCHAR (nullable)	Role or designation
consent_flag	BOOLEAN (required)	Must be TRUE to store record
utm_source	VARCHAR (nullable)	Event tracking tag from QR code URL
status	ENUM (NEW/CONTACTED/ARCHIVED)	Admin lead status
admin_notes	TEXT (nullable)	Internal notes (admin-only)
registered_at	TIMESTAMP	Submission timestamp
updated_at	TIMESTAMP	Last modified

6.3 lead_course_selections

Field	Type / Constraint	Description
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lead_id	FK → event_leads.id	Reference to lead
course_id	FK → event_courses.id	Reference to course
selected_at	TIMESTAMP	When selection was made

Composite unique key: (lead_id, course_id)

6.4 event_settings (Extends or Supplements profile_settings)

Field	Type / Constraint	Description
id	PK (single row)	Singleton settings row
yayasan_notice_text	TEXT	Customisable YAYASAN PENERAJU notice text
registration_page_title	VARCHAR	Page title for /event-register
registration_page_tagline	TEXT	Contextual tagline shown on landing page
duplicate_cooldown_hours	INT (default: 24)	Hours before same email can re-register
updated_at	TIMESTAMP	Last updated

7. API Amendments (Extension to Section 7 of SRS v1.0)

7.1 Public Endpoints

Endpoint	Method	Description
/api/event/courses	GET	Returns all PUBLISHED + PUBLIC courses
/api/event/register	POST	Submits participant registration form
/api/event/settings	GET	Returns public-safe event settings (notice text, page title)

7.2 Admin Endpoints (Protected)

Endpoint	Method(s)	Purpose
/api/admin/event/courses	GET, POST	List and create courses
/api/admin/event/courses/{id}	PATCH, DELETE	Edit or retire a course
/api/admin/event/leads	GET	Paginated lead list with filters
/api/admin/event/leads/{id}	GET, PATCH	View or update lead status/notes
/api/admin/event/leads/export	GET	CSV export of leads
/api/admin/event/qr	POST	Generate QR code with utm_source tag
/api/admin/event/settings	GET, PATCH	Manage event page settings

All admin endpoints follow the same JWT/NextAuth authentication model defined in SRS v1.0 Section 5.1.

8. Non-Functional Requirements

8.1 Security (Extension to SRS v1.0 Section 8.1)

- HTTPS required for all public endpoints (inherited from base SRS).
- All form inputs must be server-side validated and sanitised before database write.
- Rate limiting must be applied to POST /api/event/register (max 10 submissions per IP per hour).
- Consent flag must be enforced at the API layer — registration records without consent = TRUE must be rejected.
- Admin lead export endpoint must be protected and require valid admin session.
- UTM source parameter must be sanitised and length-limited before storage.

8.2 Performance (Extension to SRS v1.0 Section 8.2)

- Registration form page must load under 2 seconds on a standard 4G mobile connection.
- Course list must be server-side rendered or statically generated with ISR (Incremental Static Regeneration) to ensure fast load even under high QR scan volume.
- CSV export must be streamed for large lead sets (> 1,000 records) to prevent server timeout.

8.3 Accessibility

- Form must be keyboard-navigable.
- All form fields must have proper labels for screen reader compatibility.
- Error messages must be clearly associated with the relevant form field (ARIA attributes).

8.4 Compliance

- Consent checkbox is mandatory before any personal data is stored (PDPA compliance, Malaysia).
- Admin must be able to delete a specific lead upon request (Right to Erasure, consistent with SRS v1.0 Section 8.4).
- Registration confirmation page must reference the trainer's contact email for data deletion requests.

9. UX Requirements

9.1 Public Registration Page

- Design must be clean, professional, and consistent with Sulfri's personal brand identity.
- YAYASAN PENERAJU notice must be visually prominent — displayed in a highlighted banner above the course list.
- Course selection cards must be visually distinct and easy to tap on mobile.
- Progress indication (e.g., 'Step 1 of 2') may be used to reduce form abandonment.
- Success page must feel rewarding and professional — not a generic blank confirmation.

9.2 Admin Lead Management

- Lead table must support column sorting (by name, date, organisation).
- Quick-action buttons must be visible inline: 'Mark as Contacted', 'View Details'.
- Export button must be clearly accessible without needing to scroll.
- Course management page must show registrant interest count next to each course as a live metric.

10. Acceptance Criteria (Extension to Section 10 of SRS v1.0)

The feature is considered complete when all of the following criteria are met:

1. Scanning the QR code on a mobile device redirects to the /event-register page within 2 seconds.
2. A participant can complete and submit the registration form including personal details, course selection, and consent.
3. The YAYASAN PENERAJU notice is prominently displayed above the course selection section.
4. A success/confirmation page is shown after submission.
5. The system prevents duplicate submission from the same email within the configured cooldown period.
6. Admin can view all submitted leads in the backoffice, with correct course selections linked.
7. Admin can filter and search leads by name, email, organisation, and selected course.
8. Admin can export all leads to CSV, including course selections.
9. Admin can add, edit, publish, and retire courses from the course catalogue.
10. Admin can generate and download a QR code (PNG and SVG) with a custom utm_source tag.
11. Malicious input (XSS, SQL injection attempts) is rejected at the API layer.
12. No performance degradation observed with 50+ leads in the system.

11. Impact on Deployment

This CR introduces minimal infrastructure changes:

- No additional third-party services required.
- No change to hosting model (Vercel-compatible, serverless-safe).

- Database migration required for 4 new tables (event_courses, event_leads, lead_course_selections, event_settings).
- Frontend route additions required: /event-register, /event-register/success, /admin/leads, /admin/courses, /admin/qr.
- A QR code generation library (e.g., qrcode or react-qr-code) must be added as a dependency.

12. Version Control Recommendation

Upon approval of CR-03:

- Update main SRS version to v1.3
- Merge this amendment into Sections 5 (Functional), 6 (Database), 7 (API), 8 (Non-Functional), and 10 (Acceptance Criteria)
- Maintain CR log in Appendix

CR ID	Target Version	Summary
CR-01	v1.1	Credly Badge Wallet & Skills Wallet Integration
CR-02	v1.2	Certificate-to-Training Expertise Linking + Visual Skill Tree
CR-03	v1.3	Event Lead Capture & YAYASAN PENERAJU Course Registration Module

13. Strategic Value Statement

This module directly supports Sulfri's event-based business development strategy by converting every physical or virtual event touchpoint into a structured, trackable lead pipeline. Key strategic benefits include:

- Lead Generation at Scale — Every event attendee who scans the QR code is captured automatically into a centralised database.
- YAYASAN PENERAJU Programme Visibility — The module explicitly positions all courses under YAYASAN PENERAJU, reinforcing Sulfri's credibility and programme alignment with a nationally recognised body.
- Zero-friction Registration — Mobile-optimised form reduces drop-off for participants registering on the spot.
- Data-Driven Follow-up — Admin visibility into which courses attract the most interest enables targeted outreach and scheduling.
- UTM Tracking — Source-tagged QR codes allow Sulfri to measure ROI per event over time.

This CR complements CR-01 (verified credentials) and CR-02 (skill tree) by adding the final conversion layer: turning profile visitors and event participants into enrolled course leads — all managed from a single admin portal.

Appendix: CR-03 Metadata

Base SRS	Sulfri Trainer Portal — Personal Brand Edition v1.0 (+ CR-01 + CR-02 where applicable)
CR ID	CR-03
Type	Functional Enhancement — Event Lead Capture & Course Registration Module
Priority	High
Target Version	v1.3 (or v1.2 + Addendum)

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